

# ER Diagram and Module Description

## ER Diagram

```
+-----+
|  user_profiles  |
+-----+
| id (PK)         |
| username        |
| followers       |
| following       |
| posts           |
| profile_photo   |
| prediction       |
| created_at      |
+-----+
      |
      | 1
      |
      | M
      |
+-----+
| prediction_history |
+-----+
| prediction_id (PK) |
| user_id (FK)       |
| result             |
| confidence_score    |
| prediction_date     |
+-----+
```

## Relationship

- One User Profile can have multiple prediction records.
- `user_id` acts as a Foreign Key in the `prediction_history` table.
- Relationship Type: **One-to-Many (1:M)**

# Module Description

## Module 1: User Input Module

### Description

This module collects social media profile information from the user.

### Functions

- Accept profile details
- Validate user input
- Send data for analysis

### Input

- Followers Count
- Following Count
- Posts Count
- Profile Photo
- Website Information

### Output

- User profile data
- 

## Module 2: Data Preprocessing Module

### Description

This module prepares the dataset before training the Machine Learning model.

### Functions

- Remove duplicate records
- Handle missing values
- Select important features
- Prepare data for model training

### Output

- Cleaned dataset
-

## Module 3: Machine Learning Module

### Description

This module trains and tests the Machine Learning model.

### Algorithm Used

- Random Forest Classifier

### Functions

- Train model
- Test model
- Generate predictions

### Output

- Trained Machine Learning model
- 

## Module 4: Fake Profile Detection Module

### Description

This module analyzes social media profile attributes and predicts whether the profile is fake or genuine.

### Functions

- Analyze profile features
- Perform classification
- Generate prediction result

### Output

- Fake Profile
  - Genuine Profile
- 

## Module 5: Database Management Module

### Description

This module stores profile details and prediction results.

## Database

- MySQL

## Tables

- user\_profiles
- prediction\_history

## Functions

- Store profile data
- Store prediction history
- Retrieve prediction records

## Output

- Saved records
- 

# Module 6: Result Display Module

## Description

This module displays the prediction result to the user.

## Functions

- Show prediction status
- Display analysis result

## Output

- Genuine User
  - Fake Profile
- 

# Module Flow

```
User Input
  ↓
Data Preprocessing
  ↓
Machine Learning Model
  ↓
Fake Profile Detection
```

↓  
Database Storage  
↓  
Result Display